

computer vision:

computer vision is a part of AI that enables computers to capture, analyze and understand images and videos just like humans.

a) OBJECTIVES:

- Image understanding.
- object detection and Recognition
- Scene understanding in Image
- Making decision from the Image.

b) APPLICATIONS:

- Healthcare
- Face recognition.
- Self driving cars

Computer vision extracts meaningful information from Images and detects objects automatically.

a) FEATURE EXTRACTION:

It is the process of extracting important characteristics like edges, corners, shapes from the image

b) Object Recognition improves computer vision by detecting and analyzing objects accurately making machines intelligent.

LOW LEVEL IMAGE PROCESSING: In this the

quality of image is improved by removing noise and enhancing contrast.

b.

mid Level

High Level

→ Image Segmentation → Object recognition
→ Feature extraction → Decision making

4. Digital image is represented as a matrix of pixels.

a) PIXEL: pixel is the smallest unit of Digital image.

b) Image Resolution Effect:

Higher Resolution effect gives clearer and detailed image.

5. a) Image sensor:

It captures light and converts into electrical signals

b) acquisition devices:

1. Scanner

2. Digital camera

6. a) Sampling: Sampling is the process of converting continuous image into a discrete pixels.

Quantization: It is the process of
Increasing color/Brightness
Pixel values.

Perspective Projection:

It projects 3D object onto the 2D image plane, to make distant objects appear clearly or smaller.

Illumination:

Provides the light to capture visible image.

Image processing:

It is the process of enhancing or modifying images.

Image processing

→ analyze image

→ enhance image

→ output is modified
Image

Computer vision

→ understands the
image

→ output is the
decision information

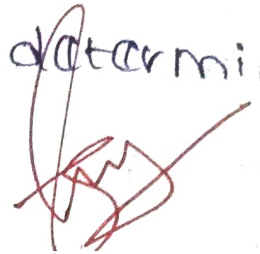
a) Healthcare:

used to detect diseases from CT, MRI and X-rays.

b) Autonomous Vehicles:

used to detect lanes, signals at traffic.

10. a) Because digital images are made up of many tiny pixels.
- b) Sampling determines the number of pixels where as resolution determines the clarity of image.

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